DEPARTMENT OF COMPUTER SCIENCE

RECORD NOTE

Record work submitted to Shri Nehru Maha Vidyalaya College of Arts and Science(Autonomous), affiliated to Bharathiar University, Coimbatore in partial fulfillment of the requirement for the Degree of

Master of Science in Computer Science

PRACTICAL III - Data Mining Lab Using R

SHRI NEHRU MAHA VIDYALAYA COLLEGE OF ARTS & SCIENCE (AUTONOMOUS)

(Affiliated to Bharathiar University)

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APRIL – 2026

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Estd. 1989 Estd. 1964

MASTER OF SCIENCE IN COMPUTER SCIENCE

This is to certify that it is a bonafide record work done by

_____Studying I year M.Sc . Computer Science

Reg .No. _____

_____ _____

Staff In – Charge **Head of the Department**

Submitted for the Autonomous Practical Examination

held on _____

_____ _____

INTERNAL EXAMINER EXTERNAL EXAMINER

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7.		Data Visualization	

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Staff In-Charge

Program No : 1
Date :

APRIORI ALGORITHM

AIM

To implement Apriori algorithm to extract association rule of data mining.

ALGORITHM

Step 1: Start the process.

Step 2: Open R Tool.

Step 3: Load required library.

Step 4: Import the dataset.

Step 5: Applying apriori() function.

Step 6: Applying inspect() function.

Step 7: Stop the process.

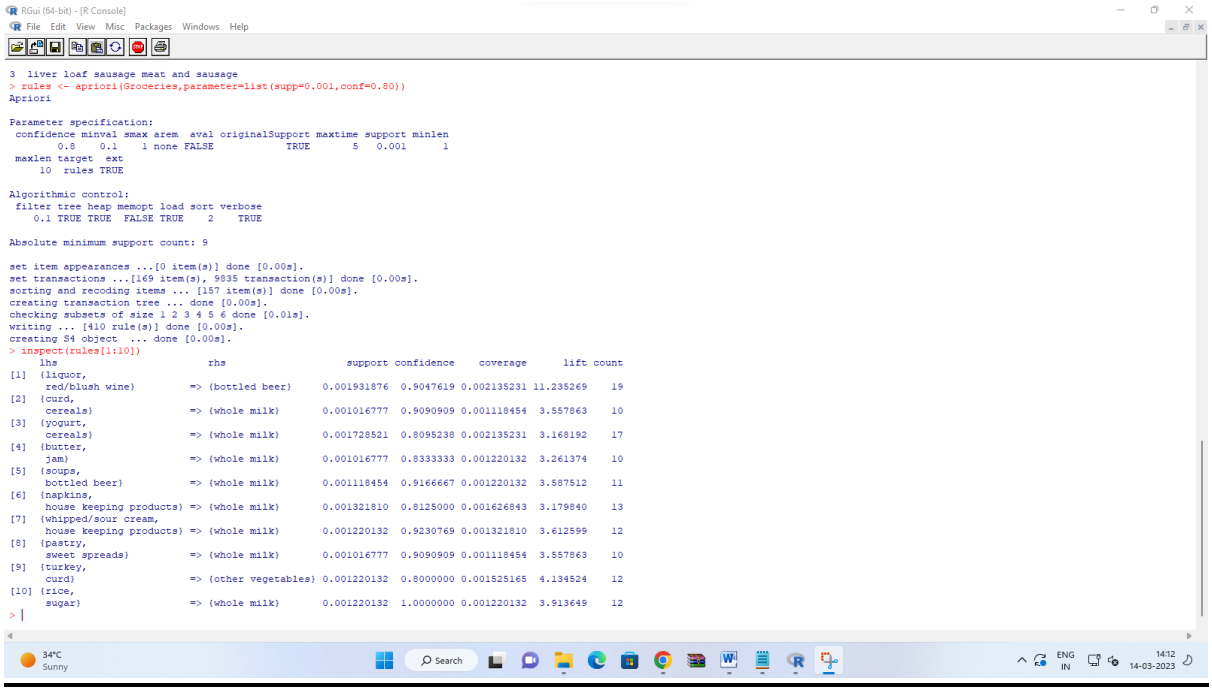
PROGRAM-1

```
install.packages("arules")
install.packages("arulesViz")
library(arules)
library(arulesViz)
data("Groceries")

# Corrected line: Use itemInfo(Groceries) instead of GroceriesitemInfo
str(Groceries)
head(itemInfo(Groceries), n=12)
summary(Groceries)

rules <- apriori(Groceries, parameter=list(supp=0.001, conf=0.80))
inspect(rules[1:10])
```

OUTPUT

```
RGui (64-bit) - [R Console]
File  Edit  View  Misc  Packages  Windows  Help

> data("Groceries")
> str(Groceries)
Formal class 'transactions' [package "arules"] with 3 slots
  ..@ data       :Formal class 'ngCMatrix' [package "Matrix"] with 5 slots
  .. .. ..@ i       : int [1:43367] 13 60 69 78 14 29 98 24 15 29 ...
  .. .. ..@ p       : int [1:9836] 0 4 7 8 12 16 21 22 27 28 ...
  .. .. ..@ Dim     : int [1:2] 169 9835
  .. .. ..@ Dimnames:List of 2
  .. .. .. ..$ : NULL
  .. .. .. ..$ : NULL
  .. .. ..@ factors : list()
  ..@ itemInfo   :'data.frame': 169 obs. of  3 variables:
  .. ..$ labels: chr [1:169] "frankfurter" "sausage" "liver loaf" "ham" ...
  .. ..$ level2: Factor w/ 55 levels "baby food","bags",..: 44 44 44 44 44 44 44 42 42 41 ...
  .. ..$ level1: Factor w/ 10 levels "canned food",..: 6 6 6 6 6 6 6 6 6 6 ...
  ..@ itemsetInfo:'data.frame': 0 obs. of  0 variables
> head(GroceriesitemInfo,n=12)
Error in h(simpleError(msg, call)) :
  error in evaluating the argument 'x' in selecting a method for function 'head': object 'GroceriesitemInfo' not found
> summary(Groceries)
transactions as itemMatrix in sparse format with
 9835 rows (elements/itemsets/transactions) and
 169 columns (items) and a density of 0.02609146

most frequent items:
       whole milk other vegetables       rolls/buns             soda
             2513             1903             1809             1715
           yogurt          (Other)
             1372            34055

element (itemset/transaction) length distribution:
sizes
   1    2    3    4    5    6    7    8    9   10   11   12   13   14   15   16
2159 1643 1299 1005  855  645  545  438  350  246  182  117   78   77   55   46
  17   18   19   20   21   22   23   24   26   27   28   29   32
  29   14   14    9   11    4    6    1    1    1    1    3    1

   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   1.000   2.000   3.000   4.409   6.000  32.000

includes extended item information - examples:
       labels  level2           level1
1 frankfurter sausage meat and sausage
2     sausage sausage meat and sausage
3  liver loaf sausage meat and sausage
> rules <- apriori(Groceries,parameter=list(supp=0.001,conf=0.80))
```

```
RGui (64-bit) - [R Console]
File  Edit  View  Misc  Packages  Windows  Help

Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[Previously saved workspace restored]

> install.packages("arules")
--- Please select a CRAN mirror for use in this session ---
trying URL 'https://mirror.niser.ac.in/cran/bin/windows/contrib/4.2/arules_1.7-5.zip'
Content type 'application/zip' length 2125554 bytes (2.0 MB)
downloaded 2.0 MB

package 'arules' successfully unpacked and MD5 sums checked

The downloaded binary packages are in
        C:\Users\Student\AppData\Local\Temp\Rtmp4C5Uce\downloaded_packages
> install.packages("arulesViz")
trying URL 'https://mirror.niser.ac.in/cran/bin/windows/contrib/4.2/arulesViz_1.5-2.zip'
Content type 'application/zip' length 1666608 bytes (1.6 MB)
downloaded 1.6 MB

package 'arulesViz' successfully unpacked and MD5 sums checked

The downloaded binary packages are in
        C:\Users\Student\AppData\Local\Temp\Rtmp4C5Uce\downloaded_packages
>
>
> library(arules)
Loading required package: Matrix

Attaching package: 'arules'

The following objects are masked from 'package:base':

    abbreviate, write

> library(arulesViz)
> data("Groceries")
> str(Groceries)
Formal class 'transactions' [package "arules"] with 3 slots
  ..@ data       :Formal class 'ngCMatrix' [package "Matrix"] with 5 slots
  .. .. ..@ i       : int [1:43367] 13 60 69 78 14 29 98 24 15 29 ...
  .. .. ..@ p       : int [1:9836] 0 4 7 8 12 16 21 22 27 28 ...
```



```
3  liver loaf sausage meat and sausage
> rules <- apriori(Groceries,parameter=list(supp=0.001,conf=0.80))
Apriori

Parameter specification:
 confidence minval smax arem  aval originalSupport maxtime support minlen
        0.8    0.1    1 none FALSE            TRUE       5   0.001      1
 maxlen target  ext
     10  rules TRUE

Algorithmic control:
 filter tree heap memopt load sort verbose
    0.1 TRUE TRUE  FALSE TRUE    2    TRUE

Absolute minimum support count: 9

set item appearances ...[0 item(s)] done [0.00s].
set transactions ...[169 item(s), 9835 transaction(s)] done [0.00s].
sorting and recoding items ... [157 item(s)] done [0.00s].
creating transaction tree ... done [0.00s].
checking subsets of size 1 2 3 4 5 6 done [0.01s].
writing ... [410 rule(s)] done [0.00s].
creating S4 object  ... done [0.00s].
> inspect(rules[1:10])
     lhs                          rhs                    support confidence    coverage      lift count
[1]  {liquor,
      red/blush wine}          => {bottled beer}     0.001931876  0.9047619 0.002135231 11.235269    19
[2]  {curd,
      cereals}                 => {whole milk}       0.001016777  0.9090909 0.001118454  3.557863    10
[3]  {yogurt,
      cereals}                 => {whole milk}       0.001728521  0.8095238 0.002135231  3.168192    17
[4]  {butter,
      jam}                     => {whole milk}       0.001016777  0.8333333 0.001220132  3.261374    10
[5]  {soups,
      bottled beer}            => {whole milk}       0.001118454  0.9166667 0.001220132  3.587512    11
[6]  {napkins,
      house keeping products} => {whole milk}       0.001321810  0.8125000 0.001626843  3.179840    13
[7]  {whipped/sour cream,
      house keeping products} => {whole milk}       0.001220132  0.9230769 0.001321810  3.612599    12
[8]  {pastry,
      sweet spreads}           => {whole milk}       0.001016777  0.9090909 0.001118454  3.557863    10
[9]  {turkey,
      curd}                    => {other vegetables} 0.001220132  0.8000000 0.001525165  4.134524    12
[10] {rice,
      sugar}                   => {whole milk}       0.001220132  1.0000000 0.001220132  3.913649    12
>
```

RESULT:

Thus, the above program was executed and the output is verified successfully.

Program No : 2
Date :

K-MEANS CLUSTERING TECHNIQUE

AIM

To implement KMeans clustering using the R tools.

ALGORITHM

Step 1: Start the process.

Step 2: Open R Tool.

Step 3: Load required library.

Step 4: Import the dataset.

Step 5: Display the summary of attitude.

Step 6: plot the graph for Kmeans result with two clusters.

Step 7: Stop the process.

PROGRAM-2

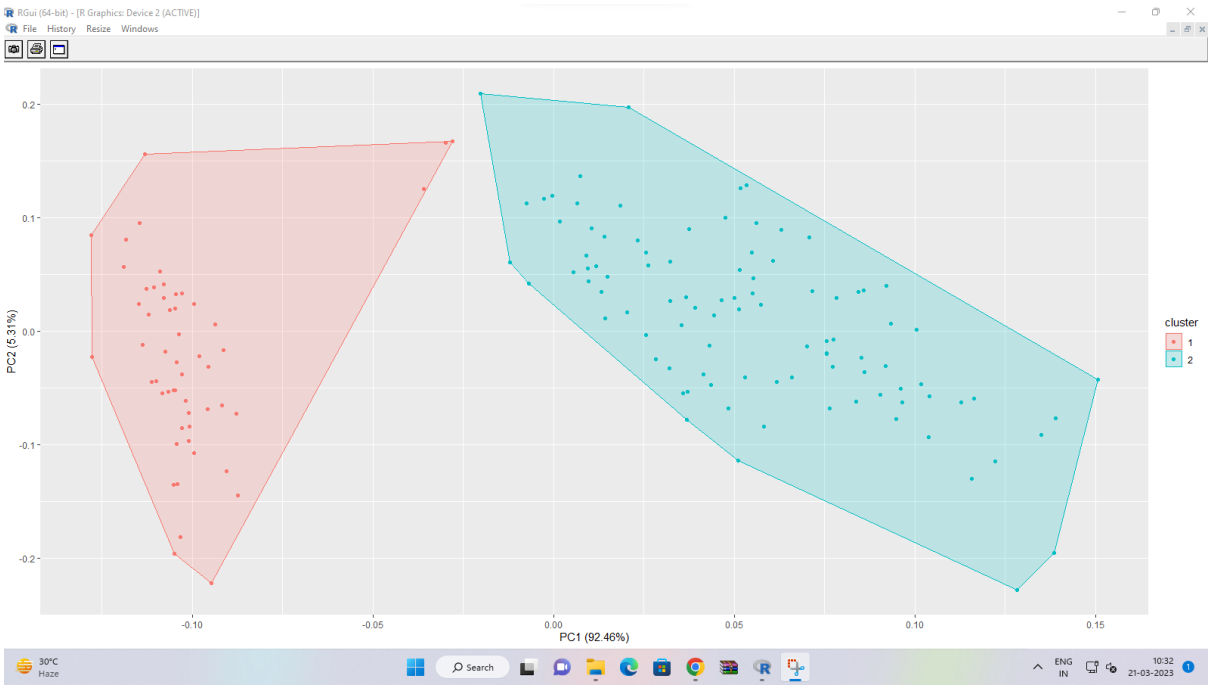
```
install.packages("dplyr")
install.packages("ggplot2")
install.packages("ggfortify")
library("ggplot2")
library("dplyr")
library("ggfortify")
summary(iris)
head(iris)
data <- select(iris, c(1:4))
kmean <- kmeans(data, 2)
kmean$centers
autoplot(kmean, data, frame = TRUE)
```

OUTPUT

```
RGui (64-bit) - [R Console]
File  Edit  View  Misc  Packages  Windows  Help

[Previously saved workspace restored]

> install.packages("dplyr")
--- Please select a CRAN mirror for use in this session ---
trying URL 'https://mirror.niser.ac.in/cran/bin/windows/contrib/4.2/dplyr_1.1.0.zip'
Content type 'application/zip' length 1548314 bytes (1.5 MB)
downloaded 1.5 MB

package 'dplyr' successfully unpacked and MD5 sums checked
Warning: cannot remove prior installation of package 'dplyr'
Warning: restored 'dplyr'

The downloaded binary packages are in
        C:\Users\Student\AppData\Local\Temp\Rtmpghaba5\downloaded_packages
Warning message:
In file.copy(savedcopy, lib, recursive = TRUE) :
  problem copying C:\Users\Student\AppData\Local\Programs\R\R-4.2.2\library\00LOCK\dplyr\libs\x64\dplyr.dll to C:\Users\Student\AppData\Local\Programs\R\R-4.2.2\library\dplyr\libs\x64\dplyr.dll: $
> install.packages("ggplot2")
trying URL 'https://mirror.niser.ac.in/cran/bin/windows/contrib/4.2/ggplot2_3.4.1.zip'
Content type 'application/zip' length 4223652 bytes (4.0 MB)
downloaded 4.0 MB

package 'ggplot2' successfully unpacked and MD5 sums checked

The downloaded binary packages are in
        C:\Users\Student\AppData\Local\Temp\Rtmpghaba5\downloaded_packages
> install.packages("ggfortify")

  There is a binary version available but the source version is later:
         binary source needs_compilation
ggfortify 0.4.15 0.4.16             FALSE

installing the source package 'ggfortify'

trying URL 'https://mirror.niser.ac.in/cran/src/contrib/ggfortify_0.4.16.tar.gz'
Content type 'application/x-gzip' length 1768532 bytes (1.7 MB)
downloaded 1.7 MB

* installing *source* package 'ggfortify' ...
** package 'ggfortify' successfully unpacked and MD5 sums checked
** using staged installation
** R
** inst
** byte-compile and prepare package for lazy loading
Warning message:
```

```
RGui (64-bit) - [R Console]
File  Edit  View  Misc  Packages  Windows  Help

  There is a binary version available but the source version is later:
         binary source needs_compilation
ggfortify 0.4.15 0.4.16             FALSE

installing the source package 'ggfortify'

trying URL 'https://mirror.niser.ac.in/cran/src/contrib/ggfortify_0.4.16.tar.gz'
Content type 'application/x-gzip' length 1768532 bytes (1.7 MB)
downloaded 1.7 MB

* installing *source* package 'ggfortify' ...
** package 'ggfortify' successfully unpacked and MD5 sums checked
** using staged installation
** R
** inst
** byte-compile and prepare package for lazy loading
Warning message:
package 'ggplot2' was built under R version 4.2.3
** help
*** installing help indices
** building package indices
** installing vignettes
** testing if installed package can be loaded from temporary location
Warning: package 'ggplot2' was built under R version 4.2.3
** testing if installed package can be loaded from final location
Warning: package 'ggplot2' was built under R version 4.2.3
** testing if installed package keeps a record of temporary installation path
* DONE (ggfortify)

The downloaded source packages are in
        'C:\Users\Student\AppData\Local\Temp\Rtmpghaba5\downloaded_packages'
> library("ggplot2")
Warning message:
package 'ggplot2' was built under R version 4.2.3
> library("dplyr")

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
> library("ggplot2")
Warning message:
package 'ggplot2' was built under R version 4.2.3
> library("dplyr")

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union

> library("ggfortify")
> summary(iris)
  Sepal.Length    Sepal.Width     Petal.Length    Petal.Width
 Min.   :4.300   Min.   :2.000   Min.   :1.000   Min.   :0.100
 1st Qu.:5.100   1st Qu.:2.800   1st Qu.:1.600   1st Qu.:0.300
 Median :5.800   Median :3.000   Median :4.350   Median :1.300
 Mean   :5.843   Mean   :3.057   Mean   :3.758   Mean   :1.199
 3rd Qu.:6.400   3rd Qu.:3.300   3rd Qu.:5.100   3rd Qu.:1.800
 Max.   :7.900   Max.   :4.400   Max.   :6.900   Max.   :2.500
       Species
 setosa    :50
 versicolor:50
 virginica :50


> head(iris)
  Sepal.Length Sepal.Width Petal.Length Petal.Width Species
1          5.1         3.5          1.4         0.2  setosa
2          4.9         3.0          1.4         0.2  setosa
3          4.7         3.2          1.3         0.2  setosa
4          4.6         3.1          1.5         0.2  setosa
5          5.0         3.6          1.4         0.2  setosa
6          5.4         3.9          1.7         0.4  setosa
> data <- select(iris, c(1:4))
> kmean <- kmeans(data, 2)
> kmean$centers
  Sepal.Length Sepal.Width Petal.Length Petal.Width
1     5.005660    3.369811     1.560377    0.290566
2     6.301031    2.886598     4.958763    1.695876
> autoplot(kmean, data, frame = TRUE)
```



RESULT:

Thus, the above program was executed and the output is verified successfully.

Program No : 3
Date :

HIERARCHICAL CLUSTERING

AIM

To implement hierarchical clustering using the R tools.

ALGORITHM

Step 1: Start the process.

Step 2: Open R Tool.

Step 3: Load required library.

Step 4: Import the dataset.

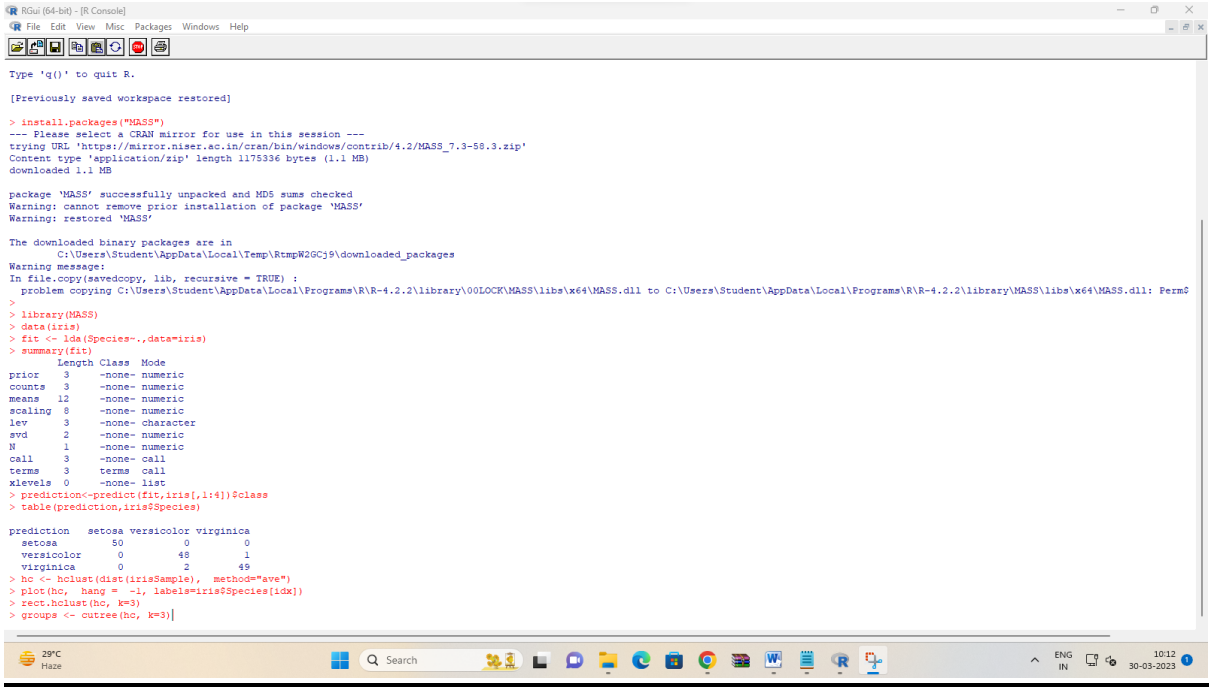
Step 5: Display the summary of attitude.

Step 6: plot the graph for Hierarchical Clustering.

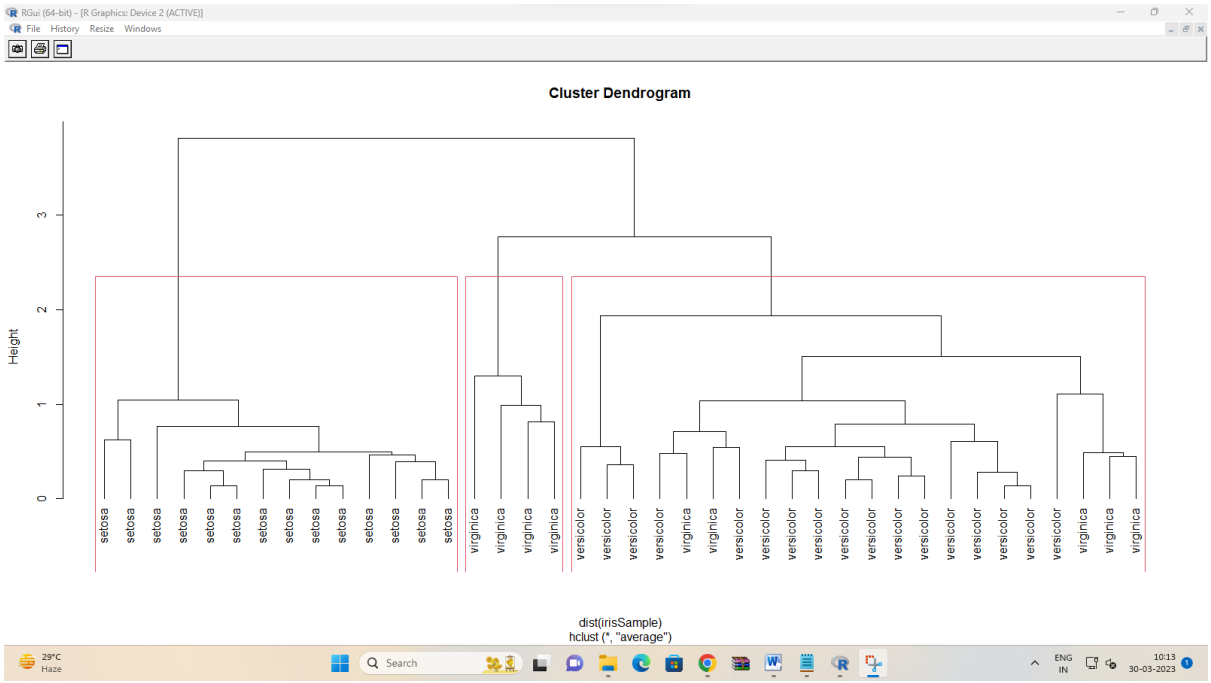
Step 7: Stop the process.

PROGRAM-3

```
# Load the MASS library
install.packages("MASS")
library(MASS)
# Load the iris dataset
data(iris)
# Perform Linear Discriminant Analysis (LDA)
# Corrected formula: Species ~ . (use all predictors)
fit <- lda(Species ~ ., data = iris)
summary(fit)
# Make predictions using the LDA model
prediction <- predict(fit, iris[, 1:4])$class
# Create a confusion table to compare predictions with actual species
table(prediction, iris$Species)
# Perform hierarchical clustering
# Use the iris dataset directly (no need for irisSample)
hc <- hclust(dist(iris[, 1:4]), method = "ave")
# Plot the dendrogram
plot(hc, hang = -1, labels = iris$Species)
rect.hclust(hc, k = 3)
# Cut the dendrogram into 3 clusters
groups <- cutree(hc, k = 3)
```



```
RGui (64-bit) - [R Console]
File  Edit  View  Misc  Packages  Windows  Help

Type 'q()' to quit R.

[Previously saved workspace restored]

> install.packages("MASS")
--- Please select a CRAN mirror for use in this session ---
trying URL 'https://mirror.niser.ac.in/cran/bin/windows/contrib/4.2/MASS_7.3-58.3.zip'
Content type 'application/zip' length 1175336 bytes (1.1 MB)
downloaded 1.1 MB

package 'MASS' successfully unpacked and MD5 sums checked
Warning: cannot remove prior installation of package 'MASS'
Warning: restored 'MASS'

The downloaded binary packages are in
        C:\Users\Student\AppData\Local\Temp\RtmpW2GCj9\downloaded_packages
Warning message:
In file.copy(savedcopy, lib, recursive = TRUE) :
  problem copying C:\Users\Student\AppData\Local\Programs\R\R-4.2.2\library\00LOCK\MASS\libs\x64\MASS.dll to C:\Users\Student\AppData\Local\Programs\R\R-4.2.2\library\MASS\libs\x64\MASS.dll: Perm$
>
> library(MASS)
> data(iris)
> fit <- lda(Species~.,data=iris)
> summary(fit)
        Length Class  Mode
prior    3     -none- numeric
counts   3     -none- numeric
means   12     -none- numeric
scaling  8     -none- numeric
lev      3     -none- character
svd      2     -none- numeric
N        1     -none- numeric
call     3     -none- call
terms    3     terms  call
xlevels  0     -none- list
> prediction<-predict(fit,iris[,1:4])$class
> table(prediction,iris$Species)

prediction   setosa versicolor virginica
  setosa         50          0         0
  versicolor      0         48         1
  virginica       0          2        49
> hc <- hclust(dist(irisSample),  method="ave")
> plot(hc,  hang =  -1, labels=iris$Species[idx])
> rect.hclust(hc, k=3)
> groups <- cutree(hc, k=3)
```



RESULT:

Thus, the above program was executed and the output is verified successfully.

Program: 4
Date:

CLASSIFICATION ALGORITHM

AIM:

To implement Classification Algorithm KNN using R Tool.

ALGORITHM:

Step 1: Start the process.

Step 2:Open the R Tool.

Step 3: Call the library function MASS.

Step 4: Select the dataset.

Step 5: Display the summary of fit.

Step 6: Display the table using Classification algorithm.

Step 7: Stop the process.

PROGRAM -4

```
install.packages("rpart")
library(rpart)
fit <- rpart(Kyphosis ~ Age+ Number+Start,method="class",data=kyphosis)
printcp(fit)
plotcp(fit)
summary(fit)
plot(fit,uniform=TRUE,main="Classification Tree for Kyphosis")
text(fit,use.n=TRUE,all=TRUE,cex=.8)
```

```
RGui (64-bit) - [R Console]
File  Edit  View  Misc  Packages  Windows  Help

  Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[Previously saved workspace restored]

> install.packages("rpart")
--- Please select a CRAN mirror for use in this session ---
trying URL 'https://mirror.niser.ac.in/cran/bin/windows/contrib/4.2/rpart_4.1.19.zip'
Content type 'application/zip' length 950269 bytes (927 KB)
downloaded 927 KB

package 'rpart' successfully unpacked and MD5 sums checked
Warning: cannot remove prior installation of package 'rpart'
Warning: restored 'rpart'

The downloaded binary packages are in
        C:\Users\Student\AppData\Local\Temp\Rtmp6BRZ1k\downloaded_packages
Warning message:
In file.copy(savedcopy, lib, recursive = TRUE) :
  problem copying C:\Users\Student\AppData\Local\Programs\R\R-4.2.2\library\00LOCK\rpart\libs\x64\rpart.dll to C:\Users\Student\AppData\Local\Programs\R\R-4.2.2\library\rpart\libs\x64\rpart.dll: $
>
> library(rpart)
> fit <- rpart(Kyphosis ~ Age+ Number+Start,method="class",data=kyphosis)
> printcp(fit)

Classification tree:
rpart(formula = Kyphosis ~ Age + Number + Start, data = kyphosis,
    method = "class")

Variables actually used in tree construction:
[1] Age   Start

Root node error: 17/81 = 0.20988

n= 81

        CP nsplit rel error xerror    xstd
1 0.176471      0   1.00000 1.0000 0.21559
2 0.019608      1   0.82353 1.0000 0.21559
```

```
RGui (64-bit) - [R Console]
File  Edit  View  Misc  Packages  Windows  Help

1 0.176471      0   1.00000 1.0000 0.21559
2 0.019608      1   0.82353 1.0000 0.21559
3 0.010000      4   0.76471 1.0588 0.22010
> plotcp(fit)
> summary(fit)
Call:
rpart(formula = Kyphosis ~ Age + Number + Start, data = kyphosis,
    method = "class")
  n= 81

          CP nsplit rel error   xerror      xstd
1 0.17647059      0 1.0000000 1.000000 0.2155872
2 0.01960784      1 0.8235294 1.000000 0.2155872
3 0.01000000      4 0.7647059 1.058824 0.2200975

Variable importance
 Start    Age Number
    64     24     12

Node number 1: 81 observations,    complexity param=0.1764706
  predicted class=absent   expected loss=0.2098765  P(node) =1
    class counts:    64    17
   probabilities: 0.790 0.210
  left son=2 (62 obs) right son=3 (19 obs)
  Primary splits:
      Start  < 8.5  to the right, improve=6.762330, (0 missing)
      Number < 5.5  to the left,  improve=2.866795, (0 missing)
      Age    < 39.5 to the left,  improve=2.250212, (0 missing)
  Surrogate splits:
      Number < 6.5  to the left,  agree=0.802, adj=0.158, (0 split)

Node number 2: 62 observations,    complexity param=0.01960784
  predicted class=absent   expected loss=0.09677419  P(node) =0.7654321
    class counts:    56     6
   probabilities: 0.903 0.097
  left son=4 (29 obs) right son=5 (33 obs)
  Primary splits:
      Start  < 14.5 to the right, improve=1.0205280, (0 missing)
      Age    < 55   to the left,  improve=0.6848635, (0 missing)
      Number < 4.5  to the left,  improve=0.2975332, (0 missing)
  Surrogate splits:
      Number < 3.5  to the left,  agree=0.645, adj=0.241, (0 split)
      Age    < 16   to the left,  agree=0.597, adj=0.138, (0 split)

Node number 3: 19 observations
  predicted class=present  expected loss=0.4210526  P(node) =0.2345679
```

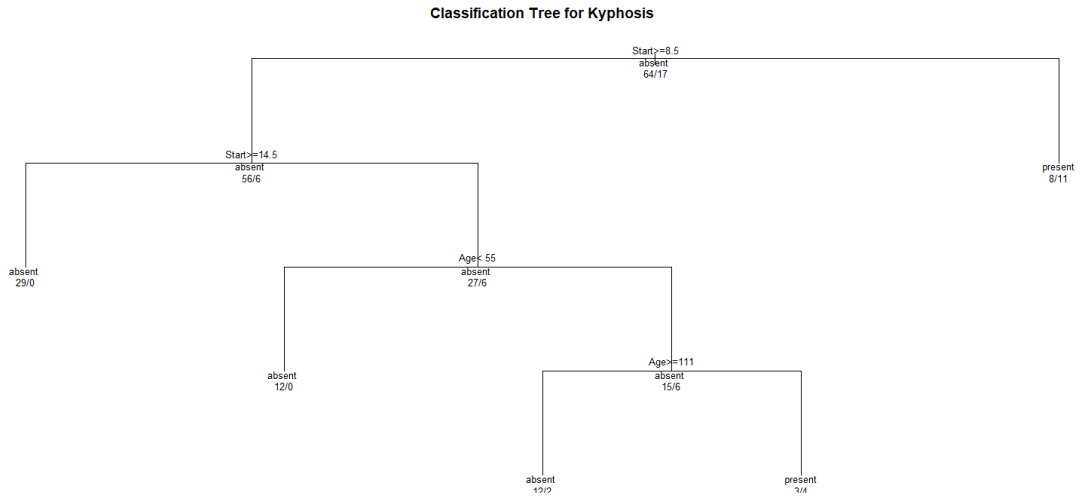
RGui (64-bit) - [R Console]

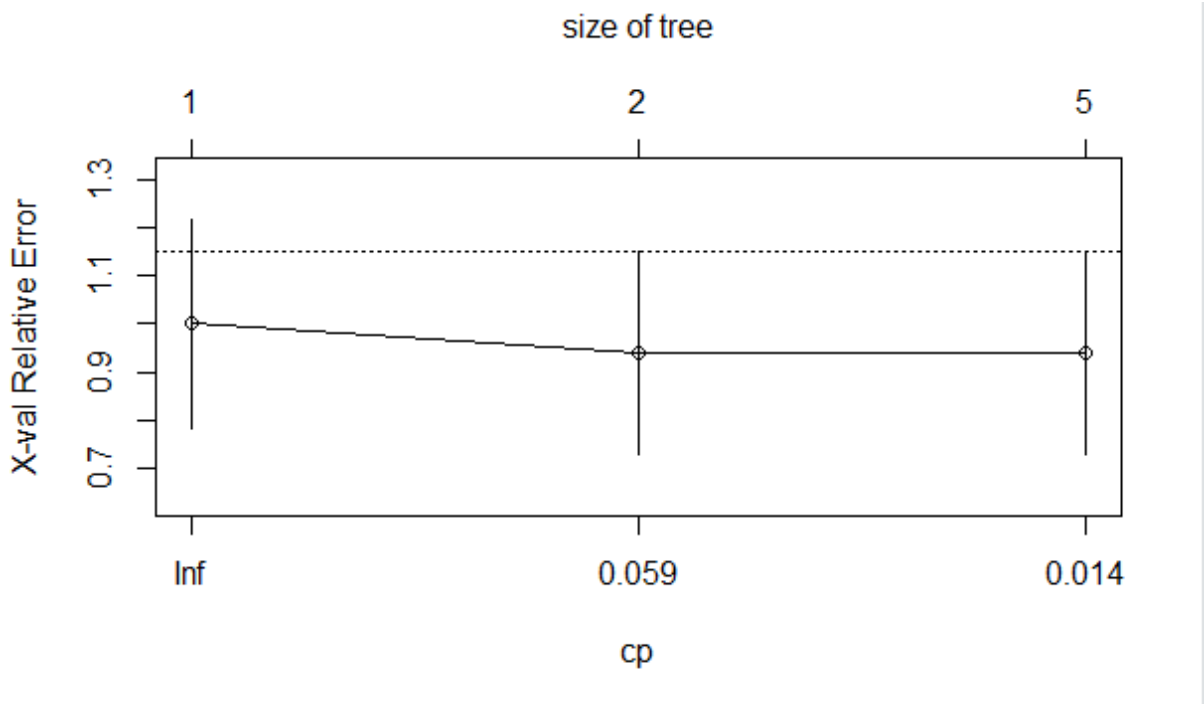
File Edit View Misc Packages Windows Help

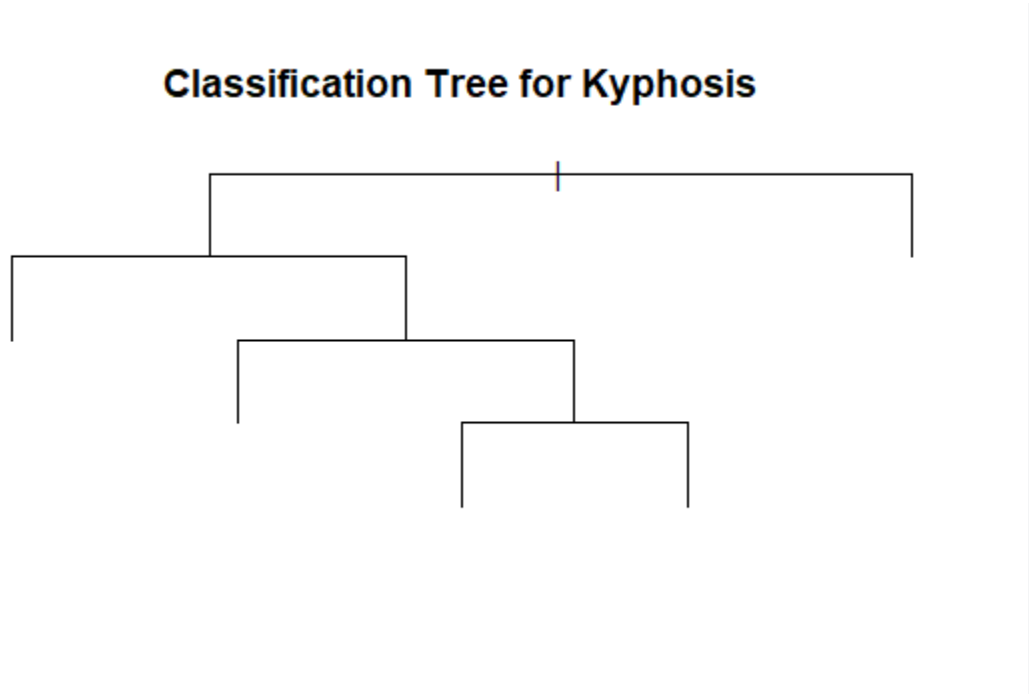
```
Node number 4: 29 observations
  predicted class=absent   expected loss=0  P(node) =0.3580247
    class counts:    29     0
   probabilities: 1.000 0.000

Node number 5: 33 observations,    complexity param=0.01960784
  predicted class=absent   expected loss=0.1818182  P(node) =0.4074074
    class counts:    27     6
   probabilities: 0.818 0.182
  left son=10 (12 obs) right son=11 (21 obs)
  Primary splits:
      Age    < 55   to the left,  improve=1.2467530, (0 missing)
      Start  < 12.5 to the right, improve=0.2887701, (0 missing)
      Number < 3.5  to the right, improve=0.1753247, (0 missing)
  Surrogate splits:
      Start  < 9.5  to the left,  agree=0.758, adj=0.333, (0 split)
      Number < 5.5  to the right, agree=0.697, adj=0.167, (0 split)

Node number 10: 12 observations
  predicted class=absent   expected loss=0  P(node) =0.1481481
    class counts:    12     0
   probabilities: 1.000 0.000

Node number 11: 21 observations,    complexity param=0.01960784
  predicted class=absent   expected loss=0.2857143  P(node) =0.2592593
    class counts:    15     6
   probabilities: 0.714 0.286
  left son=22 (14 obs) right son=23 (7 obs)
  Primary splits:
      Age    < 111  to the right, improve=1.71428600, (0 missing)
      Start  < 12.5 to the right, improve=0.79365080, (0 missing)
      Number < 3.5  to the right, improve=0.07142857, (0 missing)

Node number 22: 14 observations
  predicted class=absent   expected loss=0.1428571  P(node) =0.1728395
    class counts:    12     2
   probabilities: 0.857 0.143

Node number 23: 7 observations
  predicted class=present  expected loss=0.4285714  P(node) =0.08641975
    class counts:     3     4
   probabilities: 0.429 0.571

> plot(fit,uniform=TRUE,main="Classification Tree for Kyphosis")
> text(fit,use.n=TRUE,all=TRUE,cex=.8)
```

RGui (64-bit) - [R Graphics: Device 2 (ACTIVE)]

File History Resize Windows







RESULT:

Thus, the above program was executed and the output is verified successfully.

Program: 5
Date:

DECISION TREE

AIM:

To implement the Decision tree using R tool.

ALGORITHM:

Step 1: Start the process.

Step 2: Open the R tool.

Step 3: Call library function rpart

Step 4: Print cp.

Step 5: Plot the Decision Tree.

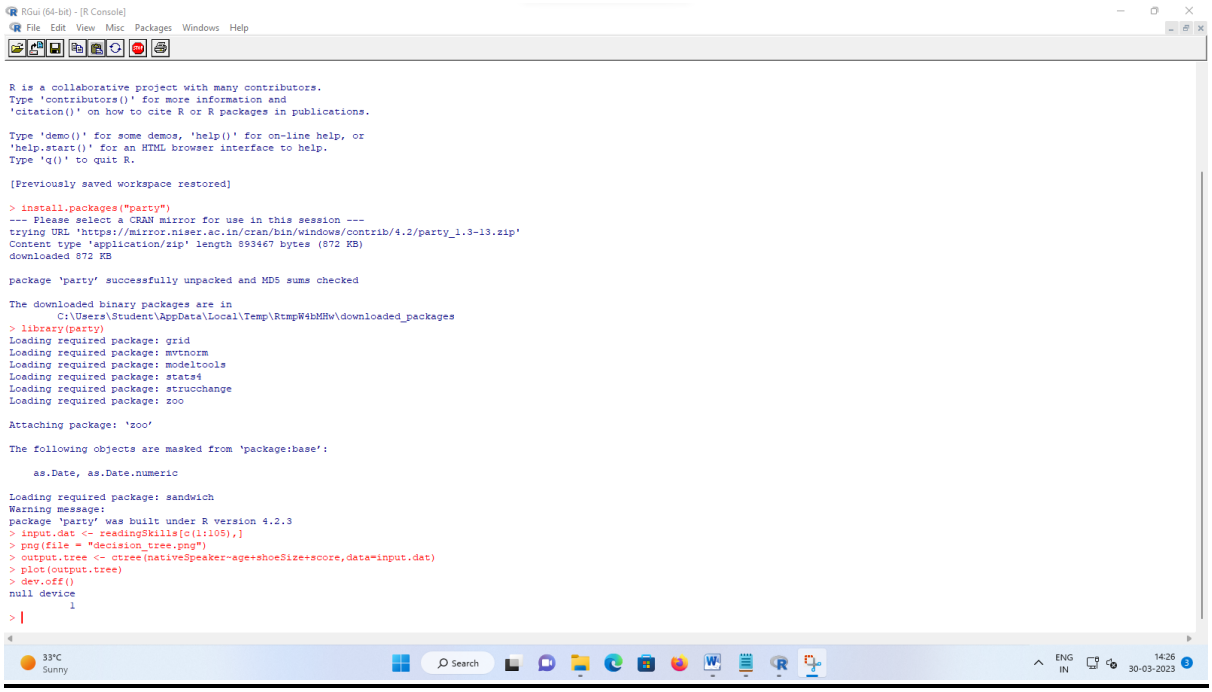
Step 6: Print the Summary.

Step 7: Stop the process.

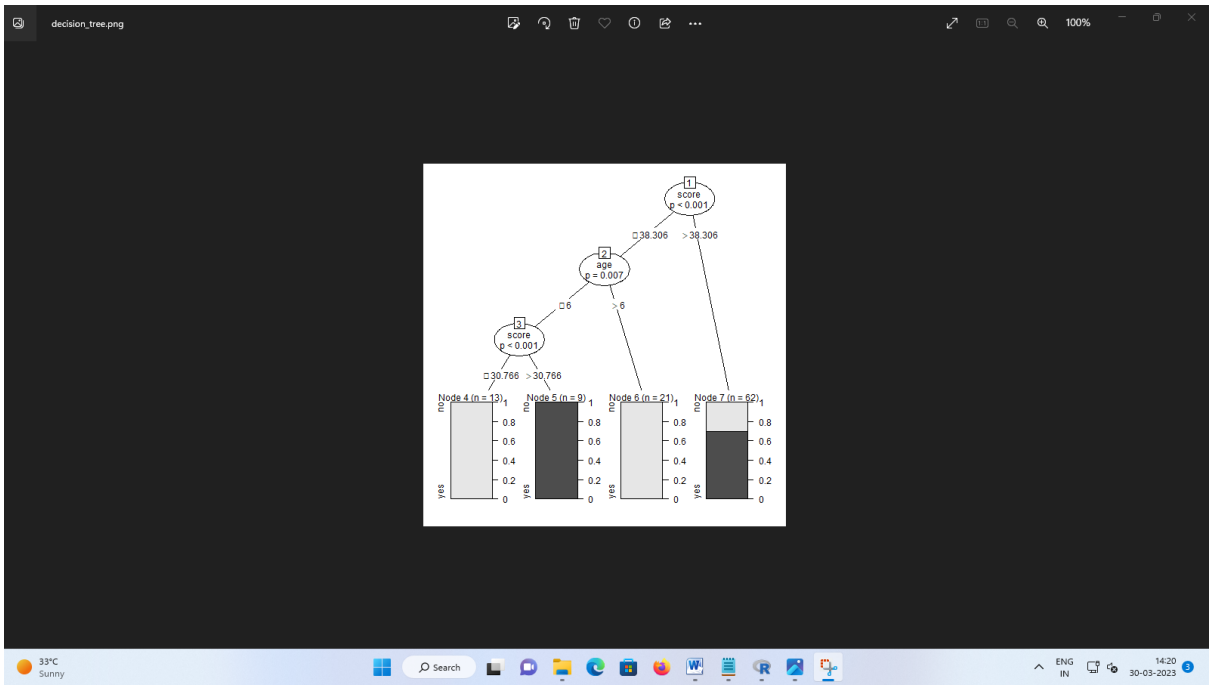
PROGRAM-5

```
install.packages("party")
library(party)
input.dat <- readingSkills[c(1:105),]
png(file = "decision_tree.png")
output.tree <- ctree(nativeSpeaker~age+shoeSize+score,data=input.dat)
plot(output.tree)
dev.off()
```

OUTPUT



```
R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[Previously saved workspace restored]

> install.packages("party")
--- Please select a CRAN mirror for use in this session ---
trying URL 'https://mirror.niser.ac.in/cran/bin/windows/contrib/4.2/party_1.3-13.zip'
Content type 'application/zip' length 893467 bytes (872 KB)
downloaded 872 KB

package 'party' successfully unpacked and MD5 sums checked

The downloaded binary packages are in
        C:\Users\Student\AppData\Local\Temp\RtmpW4bMHw\downloaded_packages
> library(party)
Loading required package: grid
Loading required package: mvtnorm
Loading required package: modeltools
Loading required package: stats4
Loading required package: strucchange
Loading required package: zoo

Attaching package: 'zoo'

The following objects are masked from 'package:base':

    as.Date, as.Date.numeric

Loading required package: sandwich
Warning message:
package 'party' was built under R version 4.2.3
> input.dat <- readingSkills[c(1:105),]
> png(file = "decision_tree.png")
> output.tree <- ctree(nativeSpeaker~age+shoeSize+score,data=input.dat)
> plot(output.tree)
> dev.off()
null device
          1
>
```



RESULT:

Thus, the above program was executed and the output is verified successfully.

Program: 6
Date:

LINEAR REGRESSION

AIM:

To implement the Linear Regression by using R Tool.

ALGORITHM:

Step 1: Start the process.

Step 2: Open the R Tool.

Step 3: Open notepad and mark details and save the file as text document.

Step 4: Print the summary of marks.

Step 5: Plot the graph for mark.

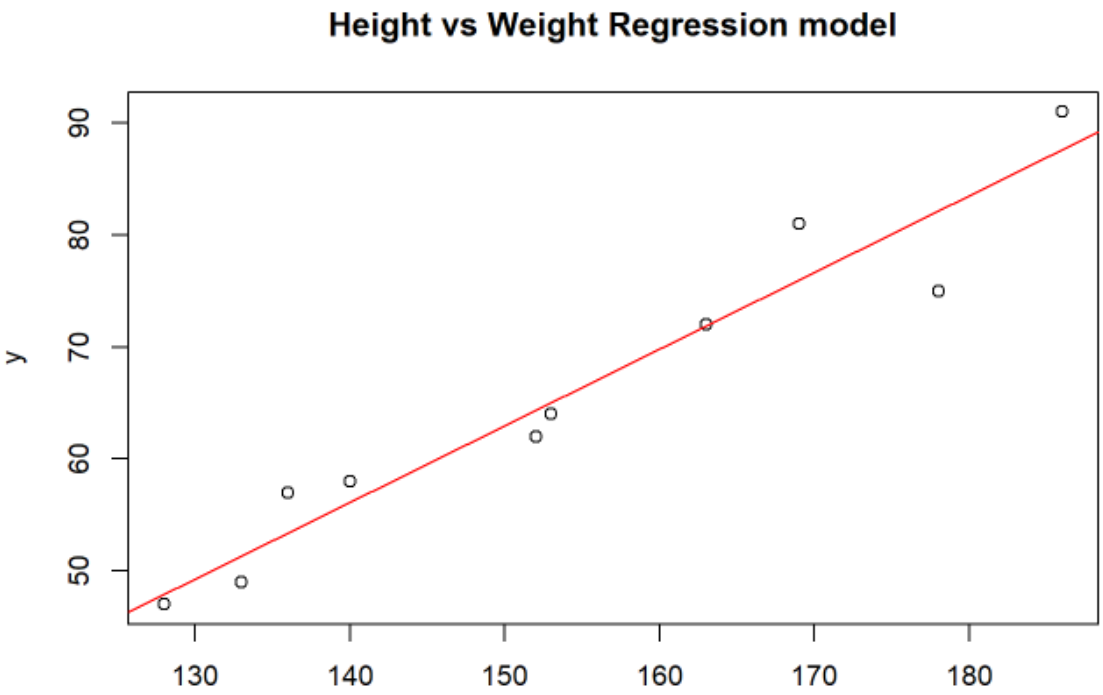
Step 6: Plot the regression chart for x & y.

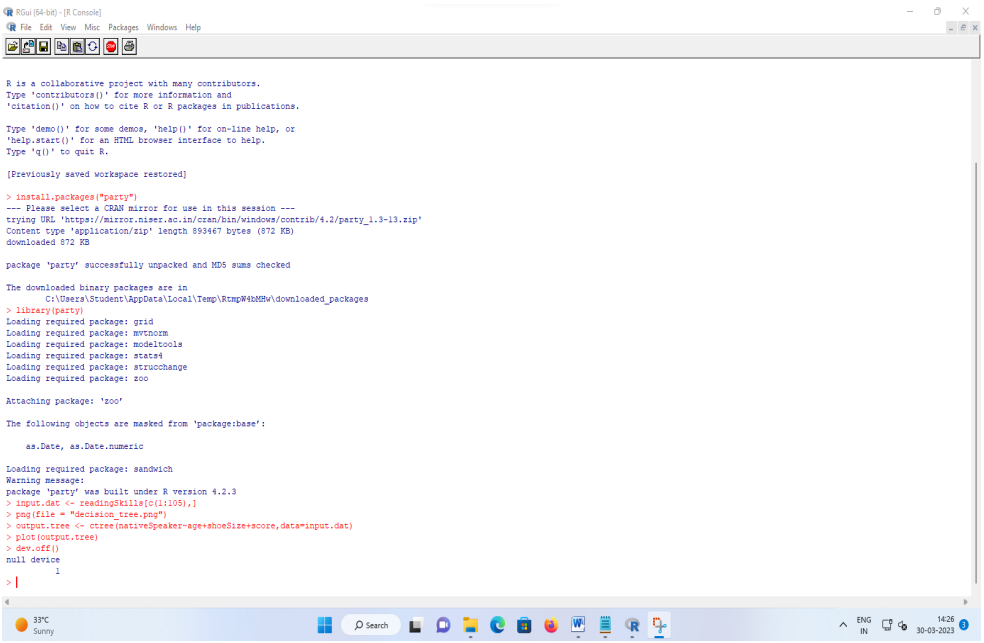
Step 7: Stop the process.

PROGRAM-6

```
x <- c(153, 169, 140, 186, 128, 136, 178, 163, 152, 133)
y <- c(64, 81, 58, 91, 47, 57,75, 72, 62, 49)
model <- lm(y~x)
print(model)
df <- data.frame(x = 182)
res <-  predict(model, df)
cat("\nPredicted value of a person with height = 182")
print(res)
png(file = "linearRegGFG.png")
plot(x, y, main = "Height vs Weight Regression model")
abline(lm(y~x))
        dev.off()
```

OUTPUT





RESULT:

Thus, the above program was executed and the output is verified successfully.

Program: 7
Date:

DATA VISUALIZATION

AIM:

To implement the Data Visualization using R Tool.

ALGORITHM:

Step 1:Start the process.

Step 2:Open the R Tool

Step 3:Given the value for slices.

Step 4:Enter the Labels.

Step 5:Specify the size of the slices.

Step 6:Use different color to different labels.

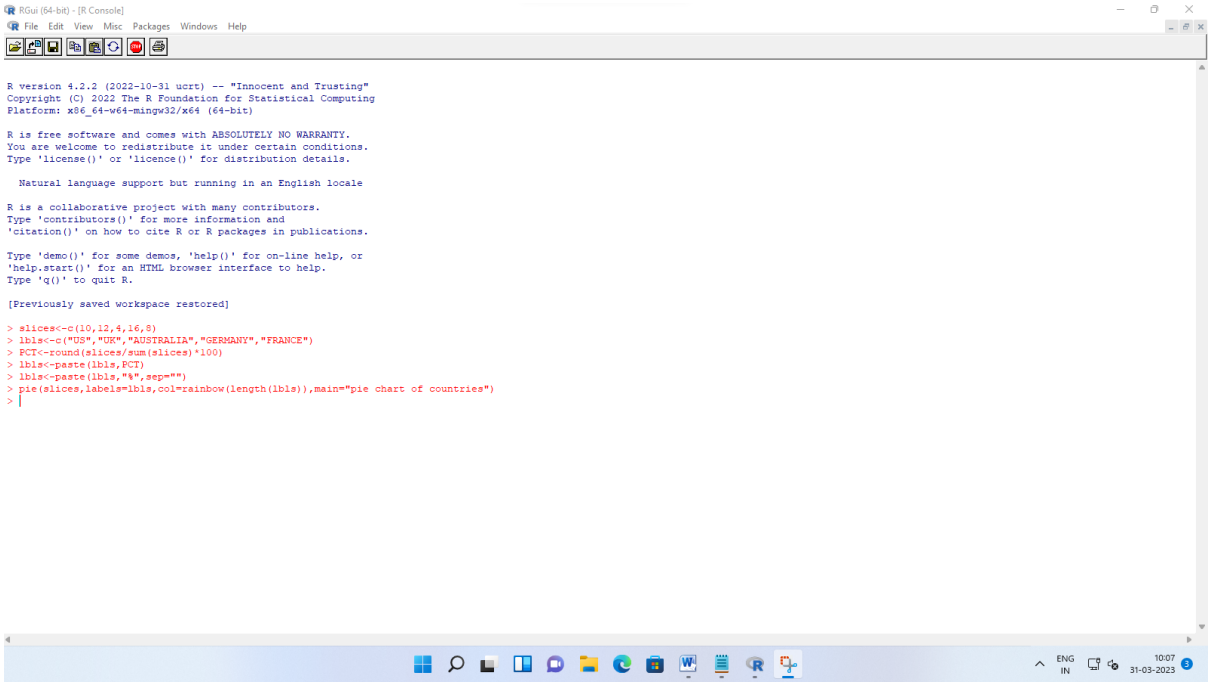
Step 7:Display Pie Chart.

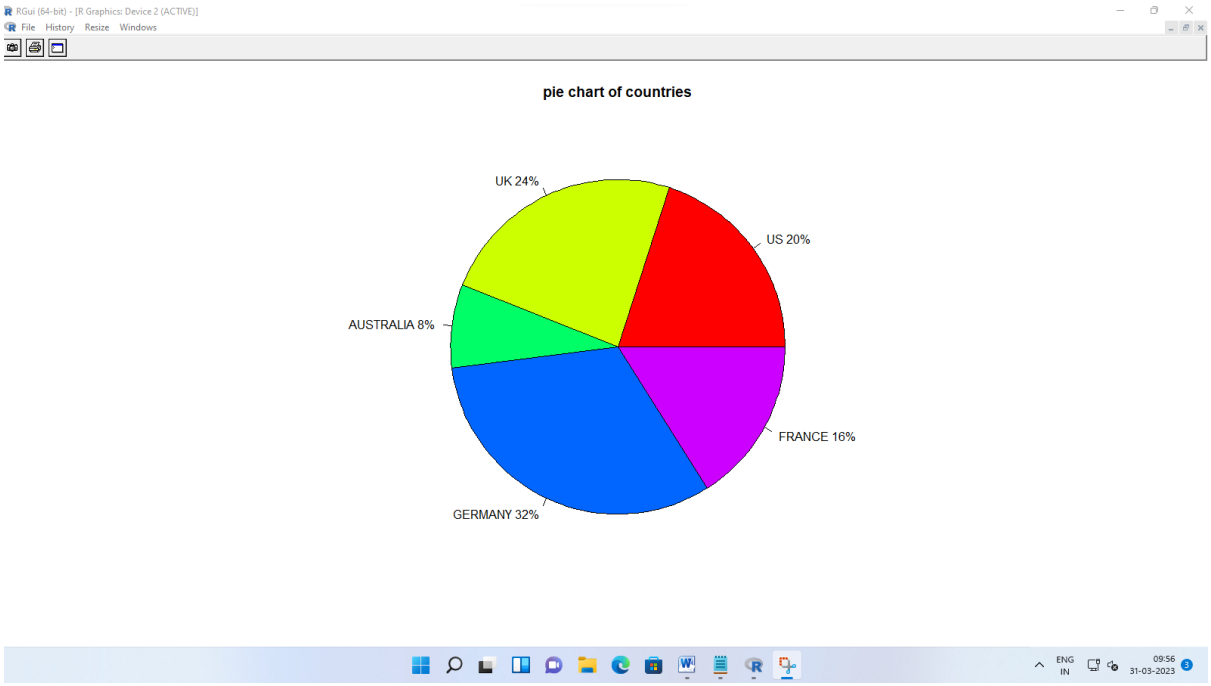
Step 8:Stop the process

PROGRAM-7

```
slices<-c(10,12,4,16,8)
lbls<-c("US","UK","AUSTRALIA","GERMANY","FRANCE")
PCT<-round(slices/sum(slices)*100)
lbls<-paste(lbls,PCT)
lbls<-paste(lbls,"%",sep="")
pie(slices,labels=lbls,col=rainbow(length(lbls)),main="pie chart of countries")
```

OUTPUT





RESULT:

Thus, the above program was executed and the output is verified successfully.